DataCORE

Parallel Tracks – Core Analysis & Multiple Regression

Afternoon: DataCORE

1. **Foundation Track — Visualization + Core Analysis** **(1:00–2:45)**
2. **Scholars Track — Multiple Regression + Interpretation** **(1:00–2:45)**
3. **Scientific Storytelling — Figure + Interpretation Pairing** **(2:45–3:00)**

*Two tracks run **in parallel** today. **Foundation** builds the core chart-and-statistic for your question; **Scholars** fits a multiple regression and interprets it carefully. Both share the same data and the same closing exercise.*

Parallel tracks — pick your lane

Foundation (RcF)

One clear chart + one key statistic, interpreted in plain language. The goal: a **defensible finding** you fully understand.

Scholars (RcS)

Multiple regression with 2+ predictors, full output table, assumption checks, and a **"holding constant"** interpretation.

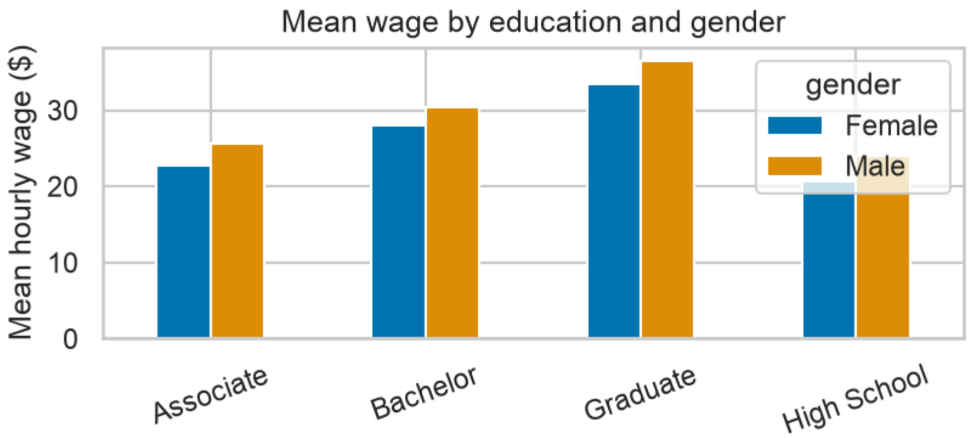
Shared dataset: the wage survey — `education`, `gender`, `region`, `experience_years`, `hourly_wage`.

1:00 – 2:45 · Foundation Track

Option A – grouped bar chart

Python R

```
tbl = wages.groupby(["education", "gender"])["hourly_wage"].mean().unstack()
tbl.plot(kind="bar", color=sns.color_palette("colorblind", 2))
plt.ylabel("Mean hourly wage ($)"); plt.xlabel("")
plt.title("Mean wage by education and gender")
plt.xticks(rotation=20); plt.legend(title="gender")
plt.tight_layout(); plt.show()
```

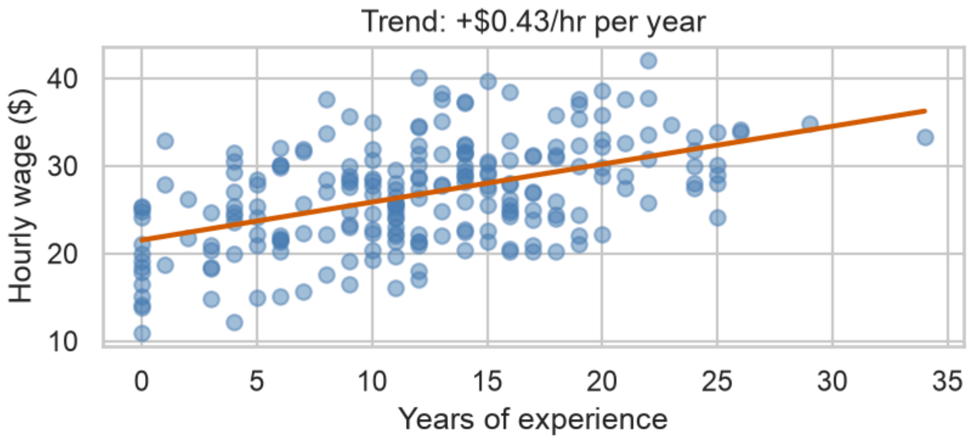


Use a grouped bar when your question compares a **numeric outcome across categories** (and sub-categories).

Option B – scatter with a trend line

Python R

```
x = wages["experience_years"]; y = wages["hourly_wage"]
m, b = np.polyfit(x, y, 1)                                   # slope, intercept
plt.scatter(x, y, alpha=0.5, color="#4a7fb5")
plt.plot(np.sort(x), b + m*np.sort(x), color="#D55E00", lw=2)
plt.xlabel("Years of experience"); plt.ylabel("Hourly wage ($)")
plt.title(f"Trend: +${m:.2f}/hr per year")
plt.tight_layout(); plt.show()
```



Use a scatter + trend line when both variables are **numeric** and you want to show a **relationship**.

🧮 One key statistic – correlation or mean difference

Python R

```
r, p = stats.pearsonr(wages["experience_years"], wages["hourly_wage"])
print(f"correlation (experience, wage): r = {r:.2f}, p = {p:.4f}")
```

```
correlation (experience, wage): r = 0.49, p = 0.0000
```

```
diff = (wages.loc[wages.gender=="Male","hourly_wage"].mean()
        - wages.loc[wages.gender=="Female","hourly_wage"].mean())
print(f"mean wage gap (Male - Female): ${diff:.2f}")
```

```
mean wage gap (Male - Female): $2.27
```

✏️ **Interpret in 2 sentences:** *"Experience and wage are positively related (r = 0.49). On average, men in this sample earned about \$2.27/hr more than women."*

AI assist — then check it yourself

Prompt the AI:

*"What does a correlation of **0.42** mean in plain English?"*

Then write your **own** one-sentence version — and compare.

A good plain-English answer:

*"A correlation of 0.42 is a **moderate positive** link: when one variable goes up, the other **tends** to go up too — but only loosely. It explains about **18%** of the variation (r^2), so much is left unexplained."*

⚠ The AI is a **drafting partner, not an authority.** Verify its claim against what you know: does "moderate" match your scatter? Is the direction right? Correlation is **not** causation.

Foundation Mission Sheet

Complete the **Session 9 Foundation Mission Sheet** on your own data.

1. **Visualize** one variable (histogram or bar).
2. **Compare** two variables (grouped bar or scatter + trend).
3. **Identify** one outlier — and ask why it's there.
4. **Write** one key insight in plain language.

45:00

1:00 – 2:45 · Scholars Track

Why **multiple** regression?

- Simple regression: one predictor (wage ~ experience).
- But wage also depends on **education**. If experienced workers are also more educated, simple regression **confounds** the two.
- Multiple regression estimates each predictor's effect **holding the others constant.**

DV = hourly wage · **IVs** = years of experience **and** education level.

wage = β_0 + β_1·experience + β_2·education

Each **β** is a *partial* slope — the effect of that variable with the rest held fixed.

Fit OLS with 2+ predictors

Python (statsmodels) | R (base)

```python
model = smf.ols("hourly_wage ~ experience_years + C(education)", data=wages).fit()
print(model.summary().tables[1])
```

```
===============================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------------------
Intercept                     19.7837      0.620     31.911      0.000      18.562      21.006
C(education)[T.Bachelor]       4.1288      0.625      6.601      0.000       2.896       5.362
C(education)[T.Graduate]       9.7785      0.814     12.010      0.000       8.174      11.383
C(education)[T.High School]   -2.5176      0.643     -3.913      0.000      -3.786      -1.250
experience_years               0.4177      0.033     12.535      0.000       0.352       0.483
===============================================================================================
```

🔍 `C(education)` / a factor adds one row **per category** (vs. a baseline). Each coefficient is that group's gap from the baseline, **holding experience constant.**

Read the full output

Python | R

```
print(f"R-squared      = {model.rsquared:.3f}")
```

```
R-squared      = 0.695
```

```
print(f"Adj. R-squared= {model.rsquared_adj:.3f}")
```

```
Adj. R-squared= 0.689
```

```
print(f"F-statistic    = {model.fvalue:.1f}    (p = {model.f_pvalue:.2e})")
```

```
F-statistic    = 122.2    (p = 3.27e-54)
```

```
print(f"N              = {int(model.nobs)}")
```

```
N              = 220
```

Each coefficient: the effect of that variable. **Std err / t / p:** is it distinguishable from 0? **R²:** variation explained. **F-stat:** is the model as a whole useful?

Say it the careful way

*"**Holding education constant,** each additional year of experience is associated with a **\$0.42** change in hourly wage (p < 0.001)."*

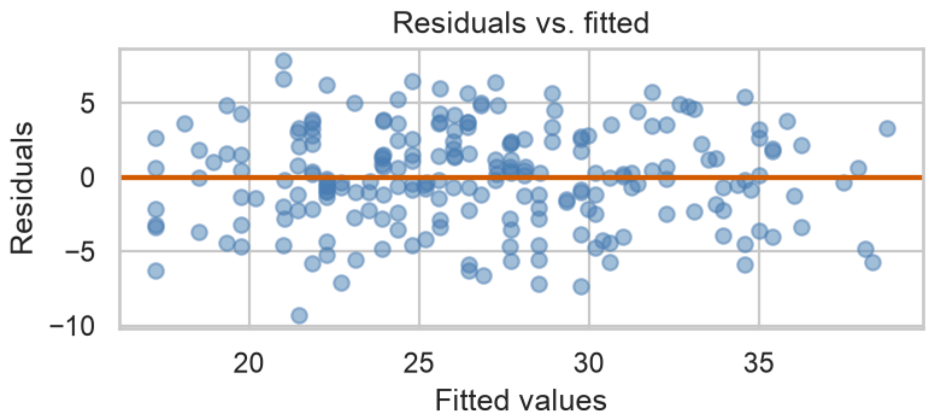
The template:
*"Holding **[controls]** constant, a one-unit increase in **[IV]** is associated with a **[coef]** unit change in **[DV]** (p = __)."*

"Associated with," not "causes."
Even with controls, this is observational — unmeasured factors may remain.

Check the assumptions

Residuals · Python | Q-Q · Python | R

```
plt.scatter(model.fittedvalues, model.resid, alpha=0.5, color="#4a7fb5")
plt.axhline(0, color="#D55E00", lw=2)
plt.xlabel("Fitted values"); plt.ylabel("Residuals")
plt.title("Residuals vs. fitted")
plt.tight_layout(); plt.show()
```



Residuals vs. fitted: want a shapeless cloud around 0 (no funnel, no curve). **Q-Q plot:** points near the line = roughly normal residuals.

Scholars Mission Sheet

Complete the **Session 9 Scholars Mission Sheet** on your own data.

1. Choose a **variable pair** (DV + key IV) plus at least one **control**.
2. State the **relationship found** — coefficient, sign, significance.
3. Write the **plain-language** "holding constant" interpretation.
4. Name **one causal limitation** — what could still confound it?

45:00

2:45 – 3:00 · Scientific Storytelling

Make them a matched pair

A great results slide is a **chart + one sentence** that fit together perfectly.

1. Write a **1-sentence interpretation** of your strongest chart.
2. Read it aloud **without** showing the chart — does it stand on its own?
3. Now show the chart — does it **prove** the sentence?

The interpretation tells the audience what to see; the chart proves it.

Example: the sentence *"Wages rise steadily with experience"* paired with a scatter that clearly trends up.

15:00